PLL Harmonic Oscillator build and calibration guide

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1. Build

This is a moderately complex but straightforward through-hole build with 4 active 100x100mm double-layer PCBs and a smaller bus PCB connecting them. The only SMD component is the L7805 +5V regulator, but it's big enough to be handled by people not used to SMD builds (like me). See section 1.4 about soldering L7805.

There's no unobtanium involved, all the components can be sourced relatively easily. The BOM lists Mouser and Thonk catalogue numbers for most of them. The most endangered species are through-hole LM13700s, but NOS can still be found and there are manufacturers making new ones. The 2k +3300ppm tempco resistor is also becoming rare but it can be found.

1.1 BOMs

The component designators use the "century" numbering system where the first digit represents the board the component is on and the next two digits are the component board id. So e.g. the first resistor on board A is R101, third cap on board B is C203 and so on. This way each component in the module has a unique designator.

The full BOM for the module is in the **module_build** folder. You can find the total quantity for each component here, and you can see which board a component is on by the century number. Also, each board build folder contains a per-board BOM which may be more helpful when stuffing individual PCBs.

1.2 PCBs

I've ordered basic FR-4 PCBs from JLCPCB. They have to be 1.6mm thick to fit into the bus board using 11mm spacers. I used HASL for boards A to E and ENIG for the panel - I wanted the GHOQ logo in gold. But HASL will work for the panel, too.

The panel could be aluminum, too, and it could be 2mm thick (standard). AFAIK silkscreen print directly on aluminum doesn't work, but it should work on solder mask.

1.3 Matching transistors, tempco resistor, thermal coupling

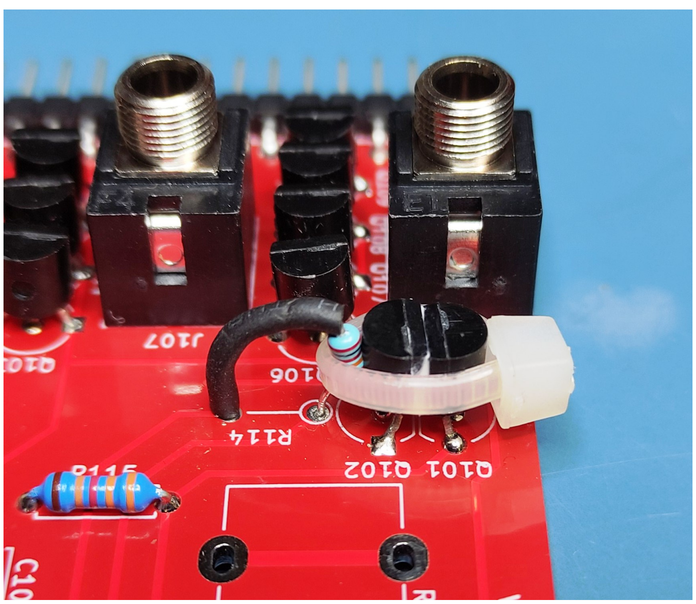
Q101 and Q102 (board A) 2N3906 transistors should be matched for equal Vbe. I've matched mine using Ian Fritz's method:

https://www.thonk.co.uk/wp-content/uploads/2016/02/ianFritz-transmat0011_144.pdf

I've also tried out a pair of unmatched transistors from the same batch and it seemed to work ok. But use a matched pair if possible.

For the R114 tempco resistor I used Ankaneohm 2k +3300ppm sourced from Thonk. I've also tried with a 1.87k +3300ppm resistor, works fine.

As for thermal coupling, Q101 and Q102 are face-to-face, with R114 standing upright beside them. Do this before soldering jacks and pots so you have enough room. First solder the transistors and put a dab of thermal paste between them. Then position the resistor with legs through the holes so it's vertical and touching Q102 but don't solder it yet. Bind the resistor and the transistors with a small ziptie and tighten it so it's a sandwich:



Squeeze it tight but don't overdo it, don't damage the enamel on the resistor. Finally solder the resistor legs.

1.4 Preset trimpots

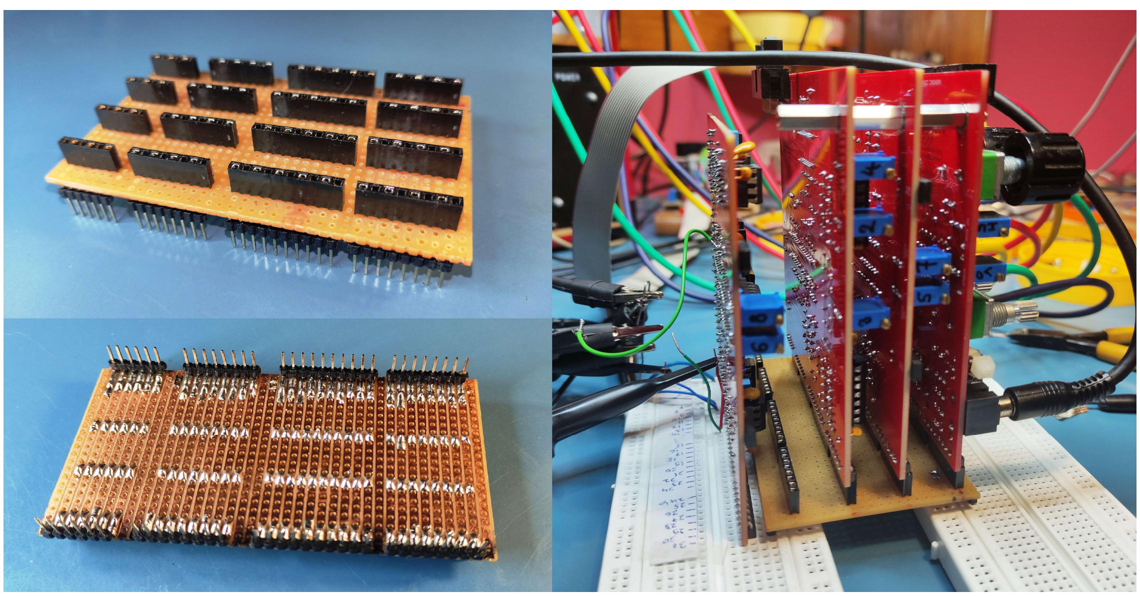
Before soldering, set all trimpot wipers to mid-position (approx.) except HF trim. Set RV105 (HF trim on board A) wiper to ground (leg 3). This will make calibration much easier.

1.5 Soldering the SMD L7805 +5V regulator

- Put some solder on both U401 leg pads (board D) and a (very) thin dab of solder on the large body pad. Use flux on everything.
- Place L7805 on the pad and solder just one leg.
- Pressing down on the IC, heat the protruding metal tab on the side and add some solder to tab edge. This should melt the dab under the body and spread the solder nicely.
- Solder the remaining leg.

1.6 Testing bus board

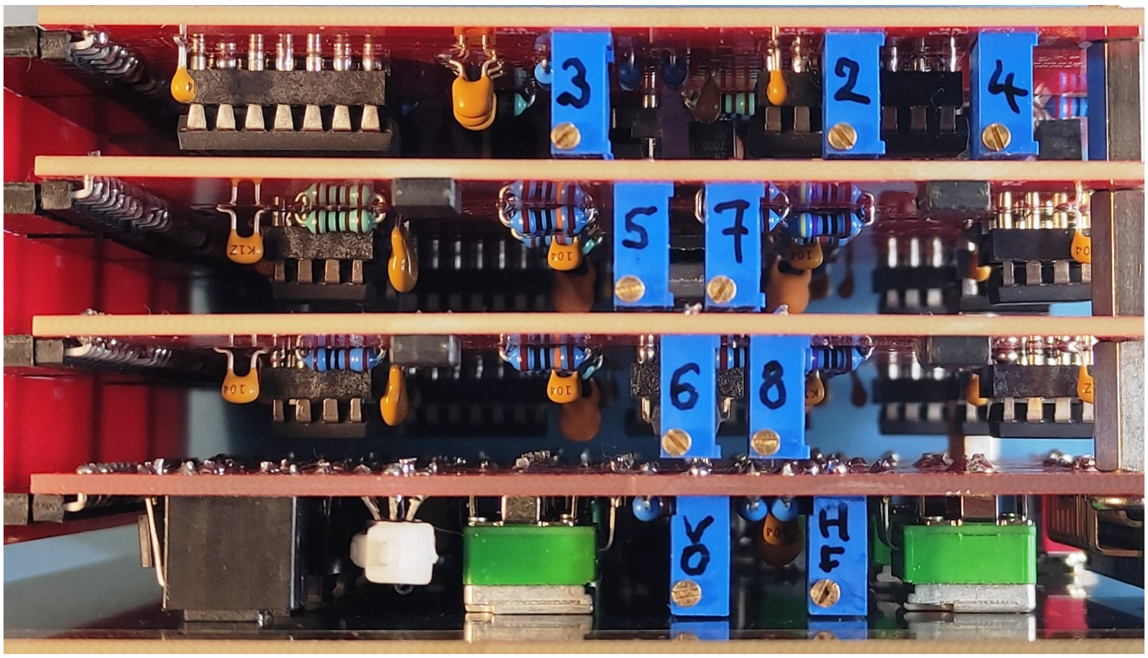
I've made a longer bus board with wider spacing between connectors for testing during the build. It's just a piece of stripboard, and I've managed to solder angled pin headers to the underside which can be plugged into 2 breadboards:



This is very useful for the build and repair if needed. The boards can be stacked in almost any order and they can be plugged into breadboards, too. This allows better access to the boards and exposes 32 test points you can probe. I'll design a test bus board PCB in KiCad and add it to the project, it's on my TODO list.

2. Calibration

The calibration can be done partly by ear, but you'll need a way to measure frequency and amplitude of the outputs to get things right. A scope is recommended. All tuning/calibration trimpots are facing up from the boards which should make the process easier.



2.1 Fundamental 1V/Oct calibration

This applies to H1, the fundamental core. 1V/Oct trim is the usual pain. Do this first, at least approximately, before calibrating harmonics. Tune octave spread with V/Oct trimpot (board A) so it's ok up to about 2kHz, then use HF trim (board A) to fix the upper range if it's flat (and it will be). But doing that will mess up V/Oct in the lower range, so you'll have to switch between the two trimpots until you decide it's good enough.

2.2 Harmonics calibration

While calibrating harmonics, make sure all modulation pots are at zero. Also, very important, make sure the PHASE MODE switch is always at the 'A' position and the PHASE pot (RV107) is always at the middle (0) position. A center-detent pot is highly recommended here.

2.2.1 H5/H6/H7/H8 tuning

These are the "real" harmonics. They may already be working (if you set H5/H6/H7/H8 wipers to mid-position), PLL error correction is very tolerant. If they're locked, you'll hear a ramp with the correct harmonic frequency. If not, you'll know. But locked or not, the cores have to be natively tuned to correct frequencies. This part can be done by ear, especially if the harmonics are already locked. But I'll describe a method where you monitor output frequencies:

First tune H1, the fundamental frequency to, say, 200Hz and keep it there. To tune H5, remove the jumper cap from H5 FB jumper (board B). This disables PLL correction and the core oscillates at its native frequency. Using the H5 TUNE trimpot (board B) tune H5 output to ~1kHz (200Hz x 5). Get within a tone or a semitone of the target frequency if you can. Then replace the jumper cap and it will lock.

Do the same thing for H6 (1.2kHz), H7 (1.4kHz) and H8 (1.6kHz), using corresponding trimpots and jumpers.

You can use any other fundamental frequency for calibration, just make sure you calculate the correct harmonic frequencies:

$$F(Hn) = F(H1) \cdot n$$

2.2.2 H2/H3/H4 slaves gain adjustment

This step requires measuring peak-to-peak amplitude of outputs.

For H2, tweak H2 GAIN trimpot (RV402 on board D) to get 10V PTP on H2 output. Do the same for H3 and H4.

Do this after H5/H6/H7/H8 tuning. If you do phase alignment, you'll have to repeat this step. Any "real" harmonic tuning change requires readjusting the corresponding slave gains.

2.2.3 H5/H6/H7/H8 phase alignment

For this you need a scope. PLLHO will work fine if you skip this step, but aligning the phase of the harmonics makes things more, well, predictable. The phase will drift anyway. To align, use the same H5/H6/H7/H8 TUNE trimpots but keep jumper caps on.

Procedure for H5: Display H1 and H5 on the scope, trigger on H1. Turn H5 TUNE trimpot (with H5 FB jumper bridged), you'll see the H5 waveform moving left or right. You're effectively detuning the core, but PLL tunes it back, changing the phase in the process. Move H5 waveform so one of the ramp resets aligns with H1 reset - pick the closest reset.

Do the same for H7, setting H7 phase alignment.

For H2/H4/H8, display the H2 waveform and align it using the H8 TUNE trim; that also aligns H4 and H8.

For H3/H6, display H3 and align using H6 TUNE trim. That aligns H6, too.

After phase alignment, repeat 2.2.2 (slave gain adjustment).

Davor Slamnig, 7/2026